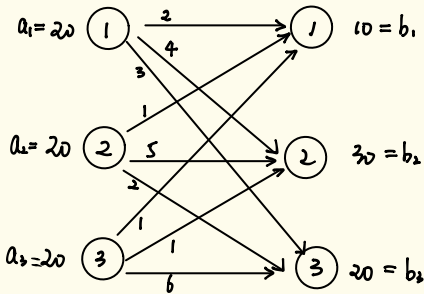


The transportation problem

Source (supply)

Destination (demand)



a_i = # units available at Source i

b_j = # units required at Destination j

c_{ij} = cost of shipping one unit from source i to destination j

Goal Design a shipping program that meets the required demands and minimizes the total shipping costs.

Note In our example, $\sum_{i=1}^m a_i = \sum_{j=1}^n b_j$, that is, the total supply = the total demand

This kind of problems are called balanced transportation problems

• LP model

Let x_{ij} be # units shipped from Source i to Destination j . The transportation problem can be formulated as a linear program.

$$\min \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$\text{s.t. } \sum_{j=1}^n x_{ij} = a_i, \quad i=1, \dots, m \quad (\text{supply constraints})$$

$$\sum_{i=1}^m x_{ij} = b_j, \quad j=1, \dots, n \quad (\text{demand constraints})$$

$$x_{ij} \geq 0 \quad \text{for all } i \text{ and } j$$

Explicitly, the equality constraints are:

$$\begin{array}{ccccccc}
 x_{11} + x_{12} + \dots + x_{1n} & & & & & & = a_1 \\
 & x_{21} + x_{22} + \dots + x_{2n} & & & & & = a_2 \\
 & & \ddots & & & & \vdots \\
 & & & x_{m1} + x_{m2} + \dots + x_{mn} & & & = a_m
 \end{array}$$

$$\begin{array}{ccccccc}
 x_{11} & & + x_{21} & & & + x_{m1} & = b_1 \\
 & x_{12} & & + x_{22} & & & + x_{m2} & = b_2 \\
 & & \ddots & & & & & \vdots \\
 & & & x_{1n} & & + x_{2n} & & + x_{mn} = b_n
 \end{array}$$

Remark 1 There are $m+n$ decision variables and $m+n$ equality constraints

2. The coefficients in the equality constraints are either 1 or 0. Each column has exactly 2 nonzero coefficients.
3. The equality constraints are not linearly independent e.g. $R_1 + R_2 + R_3 = R_4 + R_5 + R_6$ (supply = demand.)
4. There are $m+n-1$ linearly independent equality constraints. Therefore, a BFS of the transportation problem has exactly $m+n-1$ basic variables.

• The transportation tableau

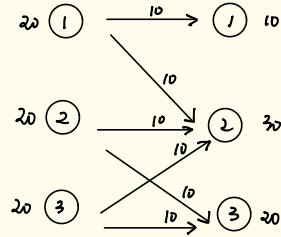
Because of the special structure, the simplex tableau of a transportation problem can be written down compactly as a transportation tableau.

	b_1	b_2	b_3	...
a_1	$\begin{array}{ c } \hline c_{11} \\ \hline \end{array}$ x_{11}	$\begin{array}{ c } \hline c_{12} \\ \hline \end{array}$ x_{12}	$\begin{array}{ c } \hline c_{13} \\ \hline \end{array}$ x_{13}	...
a_2	$\begin{array}{ c } \hline c_{21} \\ \hline \end{array}$ x_{21}	$\begin{array}{ c } \hline c_{22} \\ \hline \end{array}$ x_{22}	$\begin{array}{ c } \hline c_{23} \\ \hline \end{array}$ x_{23}	...
a_3	$\begin{array}{ c } \hline c_{31} \\ \hline \end{array}$ x_{31}	$\begin{array}{ c } \hline c_{32} \\ \hline \end{array}$ x_{32}	$\begin{array}{ c } \hline c_{33} \\ \hline \end{array}$ x_{33}	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

eg For the previous example, some feasible solutions are as below.

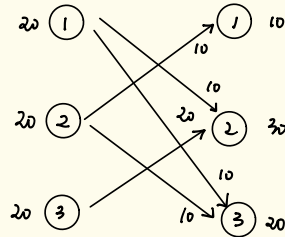
1)

	10	30	20
20	2	4	3
20	10	10	
20	1	5	2
		10	10
20	1	1	6
		10	10



2)

	10	30	20
20	2	4	3
20	1	5	2
20	10		10
20	1	1	6
		20	



Remark: 1. The empty cells mean that the corresponding variables have value 0.

2. Both examples are feasible because the values in Row i sum up to a_i for each i , the values in Column j sum up to b_j for each j , and all variable values are nonnegative.

3. The first solution is not basic and the second one is because a BFS has $m+n-1=5$ basic variables.

• Finding a BFS: the northwest corner rule.

The Northwest Corner Rule

Begin with a balanced transportation problem.

1. Examine the cell in the northwest corner of the transportation tableau. Call this cell (i, j) .
2. Assign the maximum number of units possible to cell (i, j) without violating supply constraint i and demand constraint j . Reduce the supply associated with Row i and the demand associated with Column j by the assigned amount.
3. If all units have been located, STOP. Otherwise,
 - (a) if the supply in Row i is completely satisfied, eliminate Row i ;
 - (b) if the demand in Column j is completely satisfied, eliminate Column j ;
 - (c) if both supply in Row i and Demand in Column j are completely satisfied, eliminate either Row i or Column j , but NOT both.
4. Return to Step 1.

e.g.:

	10	30	20
20	2	4	3
20	1	5	2
20	1	1	6

	10 0	30	20
10 20	2	4	3
20	1	5	2
20	1	1	6

	0	30 20	20
0 10 20	2	4	3
20	1	5	2
20	1	1	6

	10 0	30	20
10 20	2	4	3
20	1	5	2
20	1	1	6

	0	30 20	20
0 10 20	2	4	3
20	1	5	2
20	1	1	6

	0	30 20	20
0 10 20	2	4	3
20	1	5	2
20	1	1	6

	0	30 20	20
0 10 20	2	4	3
20	1	5	2
20	1	1	6

	0	30 20	20
0 10 20	2	4	3
20	1	5	2
20	1	1	6

	0	30 20	20
0 10 20	2	4	3
20	1	5	2
20	1	1	6

	0	30 20	20
0 10 20	2	4	3
20	1	5	2
20	1	1	6

A basic feasible solution:

	10	30	20
20	2	4	3
20	1	5	2
20	1	1	6

Def A loop is an ordered sequence of at least four different entries in a transportation tableau such that

- 1 Any two consecutive entries lie in either the same row or in the same column
- 2 No three consecutive entries lie in the same row or column
- 3 The last entry in the sequence has a row or column in common with the first entry in the sequence

eg

	10	30	20
20	2	4	3
	5	15	
20	1	5	2
	5		15
20	1	1	6
		15	5

	10	30	20
20	2	4	3
	5		15
20	1	5	2
	5	10	5
20	1	1	6
		20	